

EDUCATION

Vanderbilt University

B.S. in Computer Science, Minor in Neuroscience

GPA: 3.94/4.0

Expected Graduation: May 2026

TECHNICAL SKILLS

- **Languages:** Java, Python, C++, JavaScript, TypeScript, Swift, R, SQL, HTML5/CSS3
- **Frameworks & Libraries:** React, React Native, Next, Node.js, Spring Cloud, Spring Boot, Flask, Django, Spring Data JPA
- **Databases & Cloud:** MySQL, PostgreSQL, MongoDB, DynamoDB, Redis, AWS S3, Amazon SES, Google Cloud Platform
- **Machine Learning:** Pandas, NumPy, scikit-learn, PyTorch, TensorFlow, OpenCV, Spark
- **Other Tools:** Nginx, Kafka, RabbitMQ, Docker, Kubernetes, JUnit, CI/CD, Git, Postman, Linux

WORK EXPERIENCE

Software Engineer Intern

Feb 2025 – Present

Sigiriya Growth Advisors

- Built and deployed a full-stack memo generation platform, leveraging **FastAPI**, **PostgreSQL**, **Celery**, and **Redis** on the backend, **React** with **Redux** on the frontend, and deployed via **Docker** and **AWS EC2/S3**; enabled analysts to generate and manage 1,000+ investment documents with real-time task processing.
- Enhanced query performance by **designing RESTful APIs**; added composite indexing, optimized JOIN operations, and implemented connection pooling, cutting latency from 3.7s to 532 milliseconds while supporting 500+ requests/hour.

Development Intern

June 2025 – Present

VUIT - Digital Strategies & Transformation

- Engineered an end-to-end automation pipeline with **AWS Lambda**, **Microsoft Graph API**, and **SharePoint webhooks**, integrating **AI Agents** for intelligent document comparison and updates.
- Automated recurring updates to internal documentation by configuring **AWS Lambda** triggers with scheduled CloudWatch events, eliminating manual edits and improving accuracy across 200+ records.
- Migrated a 500+ page legacy website by refactoring static **HTML/CSS** into reusable **React** components, restructuring layouts, and automating redirects, improving accessibility compliance and simplifying long-term maintenance.

Software Engineer Intern

June 2024 – Aug 2024

Kongfoo Technology Company

- Developed a medical operation platform based on **Spring Boot** and **React**, integrating **RabbitMQ** and **AWS** cloud services, using **MySQL** and **MongoDB** databases.
- Utilized **Spring Boot** features including **dependency injection** to decouple controllers, services, and repositories, and **aspect-oriented programming** to implement logging, security, and transaction rollbacks; enforced consistent error handling across 10+ backend modules.
- Analyzed **MySQL** slow query logs with pt-query-digest, optimized average order query response time from 2.5s to 100ms by integrating **Redis** caching.

Software Engineer Intern

Nov 2023 – Feb 2024

Silexon AI Technology

- Developed a real-time platform for **regulatory signal monitoring and biomedical risk alerts** using **Django** and **React**, integrating **PostgreSQL** and **Redis** to support scalable querying and low-latency caching.
- Optimized critical queries by restructuring them with **Django ORM**'s annotate for in-query aggregations and subquery for nested SQL, cutting average extraction time from 5.2s to 815ms and improving risk dashboard responsiveness.
- Built high-concurrency asynchronous workflows by integrating **Celery**, **Redis**, and **Django Channels**; sustained 1,000+ tasks/sec and reduced alert latency to under 1s.

PROJECTS

Distributed Data Processing for Biomedical Image Segmentation

Oct 2024 - Present

- Conducted an open-source research project focused on biomedical image volumetric segmentation, supporting the processing of over 1,000 biomedical structures and optimizing the **segmentation algorithm**.
- Replaced **Apache Spark** with **Dask**, utilizing `Dask#DataFrame` for distributed processing and `Dask#delayed` for lazy evaluation and task scheduling, processing 500,000+ data points and improving parallel computation efficiency.
- Produced developer-facing docs with **Docusaurus** (setup, API, contribution workflow) and am first author on the paper, in preparation for submission to a peer-reviewed biomedical-imaging journal (target Q4 2025).